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Keeping an eye on the villain: Assessing the impact of surveillance cameras on crime☆

Hong Ma [a,*,1], Mingzhi Xu [b,1], Wei You [b,1], Jinmei Feng [b]

[a] Tsinghua University, Beijing, China
[b] INSE at Peking University, Beijing, China

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ABSTRACT

This study estimates the causal impact of the massive installation of surveillance cameras on crime, using novel data from China between 2014 and 2019. Leveraging the preexisting presence of local camera manufacturers as an instrument for camera deployment intensity, we find that cities with denser surveillance networks experienced significantly steeper declines in crime. The reduction is more pronounced for publicly visible crimes. Enhanced surveillance is linked to higher satisfaction with the government and a greater sense of security, which in turn leads to longer working hours. A back-of-envelope calculation shows that preventing a crime costs approximately $6,373, which is highly cost-effective.

1. Introduction

Crime control is a central responsibility of governments. A core strategy for deterring crime is the installation of public surveillance cameras. Despite significant economic costs and privacy concerns, this approach is widely implemented across governments. Several studies have estimated the local effect of public surveillance cameras on crime, typically at the neighborhood level (Burrows, 1980; King et al., 2008; Priks, 2014, 2015a; Gallagher and Fisher, 2020; Munyo and Rossi, 2020; Gómez et al., 2021). However, little is known about how the *massive* installation of surveillance cameras impacts crime rates. This relationship is important for two key reasons. First, when the surveillance network is localized, there can be a displacement effect, leading to the relocation of criminal activity from monitored to unmonitored areas (King et al., 2008; Waples et al., 2009; Cerezo, 2013; Priks, 2015b; Gómez et al., 2021). In contrast, when the surveillance network is widespread, potential offenders may find it difficult to hide from surveillance, and thus, the crime-reduction effect of unit dollar investment in surveillance cameras could vary under different operational scales. Second, on the other hand, expanding a surveillance network could result in diminishing marginal returns, as new cameras are often placed in less critical areas, making them less effective in deterring crime. Understanding whether a widespread surveillance network is economically efficient in reducing crimes and whether there are diminishing returns to investment in surveillance infrastructure is imperative for researchers and policymakers.

In this paper, we provide the first estimate of the impact of large-scale surveillance camera deployment on crime, leveraging novel data from China between 2014 and 2019. China presents a unique setting for identifying this causal relationship due to its systematic expansion of mass surveillance. Beginning in the early 2010s, the country launched nationwide initiatives such as the *Skynet*, *Smart Cities*, and *Sharp Eyes (Xueliang)* projects, aiming to construct the world's most extensive surveillance network across both urban and rural areas. By 2019,

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* Corresponding author.
E-mail addresses: mahong@sem.tsinghua.edu.cn (H. Ma), mingzhixu@nsd.pku.edu.cn (M. Xu), weiyou@nsd.pku.edu.cn (W. You), jmfeng2019@nsd.pku.edu.cn (J. Feng).

[1] Hong Ma, Mingzhi Xu, and Wei You contributed equally to this work and are recognized as co-first authors and corresponding coauthors.

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0304-3878/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

at the end of our study period, the Chinese government had installed more than 200 million cameras and invested 76.9 billion RMB (approximately 11.3 billion USD) in surveillance cameras and related equipment. This equates to an investment of 56.7 RMB (about 8.3 USD) per person.[2] Utilizing cutting-edge facial recognition technology, the system has become a critical tool for law enforcement, significantly improving the police's ability to identify suspects and solve criminal cases. The unprecedented scale of the video network, along with detailed data on its installation density and crime rates, makes China a unique setting for studying the effects of a nationwide camera surveillance network on crime rates.

To estimate this effect, we apply textual analysis techniques to granular data, generating novel measures of both crime rates and camera density. We analyze description transcripts of cases from the China Judgements Online Database (CJO) to determine crime rates by specific types of criminal activity and individual characteristics. Furthermore, we extract information on the procurement of cameras from the Chinese Government Procurement Database to measure the density of surveillance cameras. We then aggregate the document-level measures to the prefectural city level (322 cities), on which we perform our main regression analyses.[3] To further assess the broader impacts of surveillance, we integrate our data with individual-level surveys to examine how people respond to the widespread deployment of cameras.

Naturally, there is an endogeneity concern in that the deployment of surveillance cameras may be correlated with unobserved factors that also affect the crime rate. We address this concern by using the city-specific cumulative number of camera equipment producers before 2010 as an instrument for the density of surveillance cameras between 2014 and 2019. This instrument is relevant because the number of existing camera producers before 2010 strongly predicts the availability of the local supply of surveillance cameras and, consequently, the installation costs incurred by local governments when procuring cameras in subsequent years. This instrument is valid, since without the massive surveillance projects run by the central government, the size of the camera industry in a city would be independent of the crime control capabilities of the local government once the local characteristics that affect crime rates are taken into account. The size of the camera industry in 2010 impacted crime rate growth from 2014 to 2019 solely by facilitating the extensive installation of surveillance cameras after the inception of surveillance projects. We provide a set of evidence to support this key identification assumption.[4]

Our baseline IV estimation indicates that a 100 log points increase in per capita camera investment results in a 10% reduction in the crime rate. This finding holds after performing various robustness checks, including using different control variables, alternative crime data, varying measures of camera investment, multiple placebo tests, and using the number of existing electronic firms before 2010 as an alternative instrument. We also observe significant heterogeneity in the impacts across different types of crimes and regions. Surveillance cameras are particularly effective in deterring publicly visible crimes such as drug-related offenses, robbery, theft, fraud, extortion, public security breaches, and violence. Furthermore, the crime reduction effect is more pronounced in areas with higher shares of ethnic minorities and in areas with more severe gender imbalances.

In response to the increased deployment of surveillance cameras, we find that Chinese households living in cities with a higher camera density experienced a significantly greater increase in their satisfaction with the government, possibly due to the improved sense of security. These households also extended their working hours. Furthermore, we observe that women experienced a more pronounced increase in their labor supply, frequency of outdoor exercises, and satisfaction with the government, although this distinction with men is not statistically significant. These results underscore the broader positive socioeconomic effects of the extensive deployment of surveillance cameras in China.

In the final part of our analysis, we perform a back-of-the-envelope calculation of the cost of preventing a single crime by installing surveillance cameras. Here, we highlight two key findings: First, there is a significant degree of decreasing returns to the scale of the surveillance network, where the marginal costs of reducing additional crime triple as a network's scale increases from the fifth percentile to the 30th percentile of all cities. Second, under the hypothetical extreme scenario where all Chinese cities reduced their cumulative camera investment over the 2014–2019 period to the bottom 5th percentile (while cities below this threshold maintain their investment level), we estimate the cost of deterring one crime to be approximately 43,338 yuan (6373 USD), which is highly cost-effective compared to estimates in other countries. These calculations suggest the economic viability of establishing a dense, nationally interconnected public surveillance system to combat crime.

Our paper contributes to several areas of the literature. The first strand investigates the impact of surveillance cameras on crime rates. Earlier studies, such as Burrows (1980), demonstrate a significant reduction in theft and robbery after installing closed circuit televisions (CCTV) in four London Underground stations. Similarly, Priks (2015a) estimates that surveillance cameras in Stockholm's subway system effectively deterred pickpocketing and robbery. In the U.S., King et al. (2008) evaluate a San Francisco initiative where street surveillance cameras led to a roughly 20% reduction in property crime, although violent crimes were not significantly affected. In Montevideo, Uruguay, Munyo and Rossi (2020) find a similar 20% decrease in crime rates in areas monitored by cameras. These findings are supported by Gómez et al. (2021), who analyze the city of Medellín in Colombia and demonstrate the deterrent effect of surveillance technology.[5] Our research adds to this body of knowledge by establishing the first causal effect of mass surveillance on crime within the Chinese context.[6] Our findings confirm that surveillance cameras directly deter crime. Importantly, despite a high degree of decreasing returns to scale, we still

[2] For more detailed information on the installation of cameras in China, see the documentary "Amazing China" produced by China Central Television. To convert the total installation expense from RMB to USD, we use the average exchange rate from January 2019.

[3] Using prefectural cities as the unit of analysis has two advantages. First, many other datasets are available at this geographic level, allowing us to construct various control and instrumental variables. Second, and more importantly, each prefecture is a rather large administrative unit, averaging 3.93 million residents in 2010 and covering 28,038 square kilometers (with a radius of 95 km for a circle). These relatively large geographic units help alleviate concerns about the crime displacement effect, as it would be too costly for most potential offenders to travel hundreds of miles to commit a crime just to avoid cameras.

[4] To ensure robustness, we consider alternative instrumental variables. First, we construct a measure of procurement accessibility to camera producers nationwide, capturing the distance between each producer and the procuring city, akin to the widely used market access measure in regional economics. Second, we use the number of electronic producers in 2010 as an alternative instrument. As detailed in Section 4.5, our findings remain robust to these alternative specifications.

[5] Beyond surveillance cameras, the presence of police officers is also shown to deter crime significantly (Levitt, 1997; Tella and Schargrodsky, 2004; Gould and Stecklov, 2009; Draca et al., 2011a; Mello, 2019; Blesse and Diegmann, 2022). Furthermore, cognitive behavioral therapy has shown initial effectiveness in reducing crime and violence, although these effects tend to diminish with time (Blattman et al., 2017). On the other hand, social interactions with disadvantaged students in the same school could increase youth crime (Billings et al., 2019).

[6] The Chinese case is particularly important, as Gallagher and Fisher (2020) highlight that the estimated impact of surveillance cameras is highly sensitive to empirical settings. Variations in the findings across studies may reflect both differences in context and estimation method.

find that China's national surveillance network is highly cost-effective in reducing crime.

Second, our research expands the existing literature by examining how these devices affect people's subjective perceptions and labor supply behavior. On the one hand, increased security from widespread camera installation can enhance individuals' feelings of safety, thereby encouraging greater participation in public activities, as suggested by existing research (Fajnzylber et al., 2002; Goulas and Zervoyianni, 2013). On the other hand, the widespread nature of surveillance could create a climate of concern with constant oversight, potentially undermining trust in the government and reducing public engagement in civic activities (La Vigne et al., 2011; Aridor et al., 2020; Tirole, 2021). Privacy concerns related to governmental data collection have also been raised in previous studies (Priks, 2014, 2015a; Aridor et al., 2020; Tirole, 2021). Globally, governments have increasingly implemented surveillance technologies with varying levels of effectiveness and societal acceptance across different regional and national contexts (Guzik, 2016). Our study leverages a unique dataset to pinpoint the causal effects of surveillance camera deployment on individuals' subjective feelings of security. We find a notable improvement in people's feelings of security and satisfaction with their local government following the widespread installation of cameras. Our findings also indicate that a majority of Chinese citizens express a favorable attitude toward the use of surveillance cameras, probably driven by their perceived effectiveness in reducing crime. This reveals that individuals' responses to this type of surveillance are complex, but that, overall, they are influenced by its demonstrated advantages.

We also contribute to a third strand of the literature exploring the relationship between labor market performance and crime (Chalfin and McCrary, 2017). Previous research has consistently shown that declining wages and shrinking job opportunities, particularly among unskilled male workers, increase the propensity to crime (Freeman, 1996; Wilson, 1996; Raphael and Winter-Ebmer, 2001; Gould et al., 2002). In contrast, enhancing job opportunities has been linked to substantial reductions in crime rates (Burdett et al., 2003), recidivism (Schnepel, 2018), and crimes committed by immigrants (Orozco-Aleman and Gonzalez-Lozano, 2020). Dix-Carneiro et al. (2018) discover that shifts in labor market conditions mediated 75%–93% of the impact of trade liberalization shocks on crime rates. Building on these findings, our research shows how crime deterrence – as facilitated by the installation of surveillance cameras – impacts labor market outcomes. We uncover an unintended consequence of this improved perception of security: a significant increase in labor supply.

Last but not least, our findings enrich the broader literature on crime in China. Liu (2005) explores whether economically motivated crimes have escalated more rapidly than less or non-economically motivated crimes throughout the country. Xu et al. (2024) examine the relationship between the presence of "villages in the city" and crime rates. Other studies have investigated additional factors influencing crime in China, including rising male sex ratios (Edlund et al., 2013), export slowdowns (Ma et al., 2023), and widening urban–rural income disparities (Wang and Li, 2024). Our study complements these studies by directly assessing the crime-deterrent effects of surveillance cameras. Moreover, our paper introduces a novel approach to measuring both crime rates and the density of surveillance cameras, enabling a more detailed analysis of how the world's largest mass surveillance networks—*Skynet*, *Smart Cities*, *the Sharp Eyes*—contributed to crime reduction.

The remainder of the paper is organized as follows. Section 2 provides an overview of the surveillance camera system in China. Section 3 explains the construction of our dataset and presents a descriptive analysis. Section 4 outlines our empirical strategy and discusses the baseline regression results. This section also includes various robustness checks, an analysis of heterogeneous impacts, and a cost analysis of the extensive surveillance camera installation. Finally, Section 5 concludes the paper, summarizing the key insights and implications of our research.

2. Institutional background

The inception of surveillance cameras in China dates back to the 1990s. The first notable deployment, known as the "electronic police", was installed in 1997 in Shenzhen in response to frequent traffic accidents. Later, cities such as Taiyuan, Jinzhong, and Changzhi in Shaanxi province followed suit in 1999, installing cameras at major traffic intersections and establishing an accident detection and alert system.[7] Throughout the 2010s, in its quest for a safe and harmonious society, China significantly expanded its investment in surveillance technology. This expansion included the launch of several critical initiatives such as the *Skynet* Project, the *Smart Cities* Project, and the *Sharp Eyes (Xueliang)* Project. Each project is designed to expand the deployment of surveillance technology in Chinese cities, each initiative featuring distinct characteristics tailored to meet specific security and management needs. For example, the *Skynet* Project involves a nationwide network of facial recognition-enabled cameras designed to help police fight crime and identify suspects in public spaces. The *Smart Cities* Project implements a large-scale system to provide real-time warnings of natural disasters and improve crime scene analysis, shifting public governance from reactive to proactive prevention. The *Sharp Eyes* Project, although similar to the *Skynet* Project, focuses primarily on rural areas, allowing residents to monitor public spaces via television or mobile devices and directly report criminal activities to the police. This initiative significantly broadens the scope of public participation in security governance.

Fig. 1 displays the annual procurement of surveillance cameras and complementary technology by local public security agencies. Data are sourced from China's Ministry of Finance. Panel (a) presents the annual national procurement. Panel (b) shows the cumulative procurement amount. From 2002 to 2012, annual procurement shows a modest increase. Starting in 2013, procurement volumes accelerated, reaching a peak in 2018 before declining in 2019. The sample period ends in 2019 due to the COVID-19 outbreak, which significantly impacted the economy and criminal activities.[8] Overall, between 2002 and 2019, the Chinese government invested 76.9 billion RMB (approximately 11.3 billion USD, based on the average exchange rate in January 2019) in surveillance cameras and related equipment, which is equivalent to an investment of 56.7 RMB (8.3 USD) per person.

The scale and rapid expansion of China's surveillance network are unprecedented in the world. According to a documentary by China Central Television, by 2017, China had established the world's largest video surveillance network, featuring over 20 million cameras.[9] By 2019, approximately half of the world's surveillance cameras were located in China.[10] This network continues to grow: as of August 2023, there were more than 700 million surveillance cameras in China, approximately one for every two citizens, according to Kaur (2023). Several factors drive this dramatic expansion. First, advances in artificial intelligence and machine learning ("AI" for short) over the past decade have significantly impacted various industries, including public surveillance (Zuboff, 2019; Tirole, 2021; Acemoglu, 2021). China leads globally in AI research output, with facial recognition technology among the top three AI applications (Perrault et al., 2019).[11] Second,

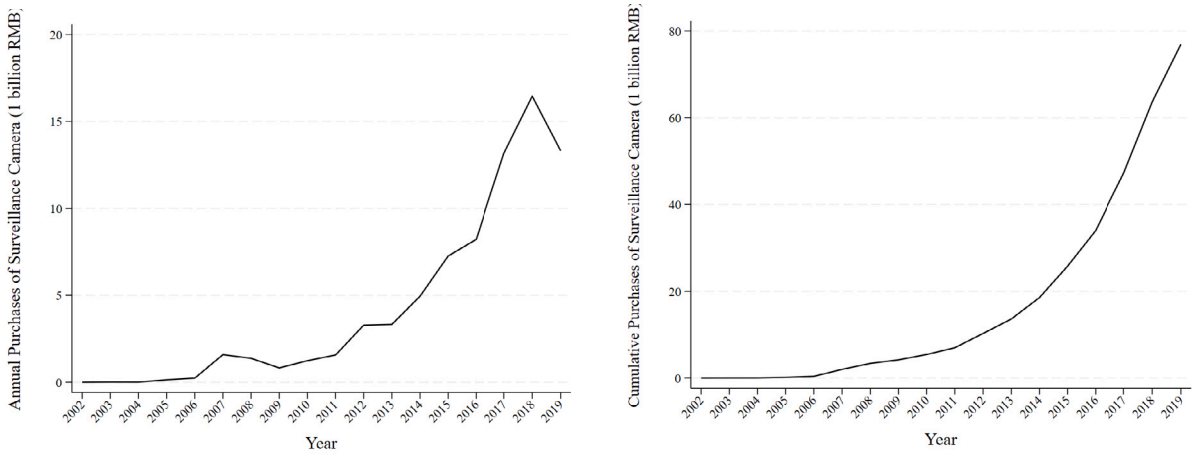
[7] For more detailed information about "electronic police", see the *Law Yearbooks of China.*

[8] The COVID-19 pandemic emerged in late 2019, leading to strict restrictions on geographic mobility that significantly reduced criminal activity. Additionally, crime rate data after 2020 are less accurately measured. For these reasons, we exclude data beyond 2020 from our analysis.

[9] See episode 5 of the documentary "Amazing China (Huihuang Zhongguo)".

[10] See Lin and Purnell (2019) (2019-12-06). "A World With a Billion Cameras Watching You Is Just Around the Corner". Wall Street Journal.

[11] For more detailed information on China's leading position in AI research, see the "China AI Development Report, 2018".



(a) Annual Procurement Volumes of Camera (b) Cumulative Procurement Volumes of Camera

Fig. 1. Time trends of camera investment. *Notes:* The left panel presents the annual trend in the procurement volumes of public surveillance cameras, while the right panel shows the cumulative amount of procurement over time.
Source: Authors' own calculation based on the government procurement data obtained from China's Ministry of Finance.

there is deep mutually reinforcing collaboration between AI firms and the Chinese government, where the government provides vast amounts of training data, essential for refining machine learning algorithms, to AI companies, which in turn utilize these data to improve algorithms and provide services to public security agencies (Beraja et al., 2023a,b). Third, the Cybersecurity Law of China, enacted in 2016, further strengthens the ability of public security and national security organizations to investigate criminal activities using data from network operators in China.

Anecdotal evidence. The efficiency of China's video surveillance network is exceptional. The system can scan the entire population of China in less than a second with an accuracy rate of 99.8%.[12] This capability was showcased in a 2017 practical test by BBC correspondent John Sudworth at the public security center in Guiyang City, Guizhou Province. The test began with Sudworth having his mugshot taken before he embarked on his "escape plan". Just seven minutes after his departure, he was identified and located by the police using the *Skynet* Project technology.[13] Another significant demonstration of its effectiveness occurred in Linyi, Shanxi Province, in 2017. A woman, while at home with her grandchildren, spotted an unlicensed vehicle on her community's surveillance feed. Seeing a suspicious individual attempting to break into her neighbor's house, she quickly used her TV's remote control to activate the emergency alert. The police responded promptly and apprehended the suspect, further highlighting the critical role of these integrated surveillance systems in crime prevention.

The anecdotal evidence strongly supports the successful deployment of surveillance systems by the government to prevent and combat crime. The most compelling evidence is sourced from reports on the official websites of the Public Security Departments, People's Courts, and People's Procuratorates in the 31 provinces of China, covering the period 2009 to 2019. We conducted searches for news reports using four specific keywords: "*Skynet* Project", "Sky Eye", "surveillance", and "cameras". Each report was manually reviewed to confirm its relevance to the use of surveillance cameras in crime prevention. This process generated two time series data sets: one tracking the number of news reports on surveillance cameras and the other focusing on the reports detailing the use of these cameras to combat crime, as shown in the Appendix Figure A.1. Consistent with the investment trends illustrated in Fig. 1, there was a notable increase in camera-related news reports starting in 2013. In particular, between 2018 and 2019, there was a significant increase in reports on camera use for crime prevention, coincident with the peak in substantial investments in surveillance technology a year earlier.

Surveillance networks have significantly aided police efforts in numerous ways, detailed in Appendix B through five representative cases. In Case 1, a notorious fugitive who had evaded capture for 20 years was inadvertently identified by a surveillance camera in a shopping center in Xiamen, Fujian Province. Despite undergoing several plastic surgeries, the camera system recognized the fugitive with 97.33% accuracy. This illustrates the pervasive reach of the video surveillance network, which leaves few hiding places for fugitives. Case 2 highlights how the surveillance system allowed police to access video footage of a crime scene and its surroundings, helping to quickly narrow down potential suspects. In Case 3, the police used evidence from the surveillance footage to confront an offender, who later confessed to the crime when faced with irrefutable evidence. Case 4 demonstrates the surveillance system's capacity to enhance police's ability to remotely visualize and position crime scenes, with an instance in which a theft victim did not realize her mobile phone was stolen until police arrested the thief and returned her phone. Finally, Case 5 underscores the effectiveness of surveillance cameras in locating missing persons. Despite various reports and anecdotal evidence supporting the efficiency of surveillance systems, a rigorous evaluation of whether their widespread installation in China has directly resulted in crime reduction has been missing. The empirical analysis in this paper fills this void, providing a detailed examination of the impact of mass surveillance on crime rates in China.

On the other hand, the adoption of private security cameras appears to be relatively limited compared to the extensive public surveillance infrastructure. Estimates suggest that between 2016 and 2018, annual household camera shipments in China averaged around 11.86 million units, while by 2018, the total stock of public surveillance cameras may have reached 200 million.[14] If we assume that all newly shipped

[12] Reported by Sohu.com on March 25, 2018, available at https://www.sohu.com/a/683530686_121687424.

[13] For more details on this test, visit https://news.cgtn.com/news/79676a4e31637a6333566d54/share_p.html.

[14] According to Liu and Shi (2021), household camera shipments reached 40.4 million units in 2020, growing at a compound annual rate of 53.5% from 2016 to 2020. Using this growth rate, we estimate the average annual shipments for 2016–2018 to be 11.86 million units. The total stock of public surveillance cameras in 2018 is based on data from https://tech.ifeng.com/a/20180504/44980719_0.shtml.

household cameras were installed, annual private cameras are roughly 5% of the total surveillance camera stock, underscoring the dominant role of government-installed cameras in crime prevention. Besides their smaller scale, private cameras are also less effective in detecting crimes that occur in public spaces, such as violent assaults, drug-related offenses, and organized criminal activity, which are the key targets of public surveillance.[15] Unlike private security systems, which are installed based on individual preferences, public camera networks are deployed strategically using crime mapping and law enforcement priorities, covering high-risk areas such as transportation hubs, streets, and commercial districts.

3. Data description and variable definition

The empirical analysis utilizes two distinct data sets to measure crime rates and the density of investment in surveillance cameras, respectively. Our study focuses specifically on prefectural cities as geographic units of the analysis.[16]

3.1. Data construction and sources

Crime rates. Our data on criminal cases are derived from the China Judgments Online Database, the official website of the Supreme People's Court (SPC) of China. This platform is legally mandated to disclose judicial documents online.[17] The CJO database encompasses a comprehensive collection of judgment documents from 2014 to 2019, excluding cases related to national security, divorce litigation, juvenile delinquency, and those resolved through mediation. The database categorizes documents into five types: criminal, administrative judgments, execution rulings, civil, and other judicial documents (such as payment orders). Each document includes the title, date, court location, and a detailed description of the case. Our study focuses on criminal offenses that violate the Criminal Law of the People's Republic of China. This focus is driven by the primary motivation behind the extensive installation of surveillance cameras: to deter criminal activities that impose significant social costs.

We perform a detailed textual analysis of the description transcripts of these cases to categorize each case into specific crime types. Specifically, we search for keywords associated with particular crimes in each judgment document and match them accordingly. We then aggregate the individual-level data to compile the total number of criminal cases by year, city, and type of crime. In our analysis, criminal cases are classified into several categories, including crimes that endanger public security, traffic felonies, violence, robbery, stealing, defraudation, extortion, drug-related offenses, prostitution, gambling, counterfeiting, intellectual property rights infringements, financial fraud, and bribery. For details, refer to Table A.3 in the appendix. For analytical convenience, we group certain types of crime into broader categories. For example, robbery, stealing, defraudation, and extortion are combined into a group termed property crime; similarly, financial and intellectual property infringement are grouped under "economic crime". Fig. 2 displays the national time trends for various criminal offenses in China from 2014 to 2019.[18]

We normalize the number of criminal cases by dividing each city's total by its 2010 resident population (per 10,000 residents) to calculate crime rates.[19] Our analysis includes a sample of 322 cities. The primary outcome variable is the growth rate of criminal cases, measured by the long difference in the logarithm of the total number of criminal cases between 2019 and 2014. In our robustness checks, we also collect crime data at the provincial level from *Procuratorial Yearbook of China* (that is, the Supreme People's Procuratorate,2009–2019). Criminal cases at the provincial level are classified based on approval of arrests or the number of prosecutions. We collect data at the provincial level to address the limitations of prefecture-level data, which is only available after 2014. Specifically, we use data from 2009–2014 to perform placebo tests.

Density of surveillance cameras. The second primary data source utilized in our study is the Chinese Government Procurement Database (CGP), maintained by China's Ministry of Finance. The CGP database provides comprehensive details on government procurements, including the date, monetary value of each contract, and information on the winners of some procurement bids. We use this database to construct measures of surveillance camera density in Chinese cities, based on the procurement contracts specifically for surveillance cameras. By conducting a textual analysis of these contracts, we extract data on the number of contracts for surveillance cameras by city-year and the price of each procurement contract. From this, we calculate the total purchases of surveillance cameras from 2014 to 2019.[20] The combined value of these contracts across 322 cities during this period amounted to approximately 61.2 billion yuan (9 billion USD). Fig. 1 illustrates the national trends in annual and cumulative purchases of surveillance cameras from 2002 to 2019. To gauge the density of surveillance cameras, we normalize the cumulative number of contracts, the cumulative number of contracts with specified prices for cameras, and the total camera purchases from 2014 to 2019 by each city's 2010 resident population (per 10,000 residents).

Instrumental variable and control variables. To construct the instrumental variables, we utilize the *Tianyancha* database, which is maintained by China's State Administration for Industry and Commerce and provides basic information (including addresses and main products) on all registered firms in China. We extract firms that produced surveillance cameras, as well as firms that produced electronic products, prior to the extensive camera procurement and installation initiatives (we choose the year 2010). This database enables us to determine the number of surveillance camera manufacturers and the number of electronic firms present in each city. Before 2010, firms producing surveillance cameras were distributed across 274 cities, with an average of 144 camera firms per city. The distribution of these firms before 2010 is illustrated in panel (d) of Fig. 3, which is used as our baseline instrument. The number of electronic firms is used as an alternative instrument. We make this choice because, as we show later, the number of existing camera firms in 2010 is a more relevant instrument for surveillance camera installation over 2014–2019. Both instruments use the number of *local* firms because municipal governments in China overwhelmingly favor local suppliers.[21] We construct other covariates that may influence the growth rate of crime using data from the 2005 and 2010 population censuses, the China City Statistical Yearbooks, and the China Economic Information Center (CEIC). Details on these data sources are provided in Appendix Table A.2.

[15] A meta-analysis by Welsh and Farrington (2009) find that surveillance systems are most effective in parking lots, reducing crime by 51%, while their impact in city centers and public housing is more limited.

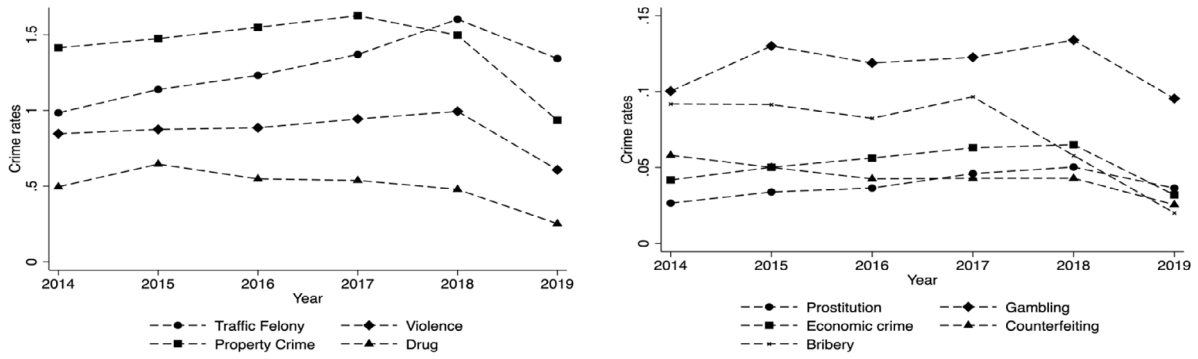
[16] China's administrative structure follows a four-tier hierarchy. Beyond the central government, the highest administrative divisions are the provincial units, which encompass provinces (e.g., Sichuan Province), autonomous regions (e.g., Tibet), and centrally-administered municipalities (e.g., Beijing, Tianjin, Shanghai, and Chongqing). Below them are prefecture-level divisions, which are further subdivided into county-level cities and counties, including autonomous counties. The lowest tier consists of townships and towns. In this paper, we use the prefectural city as the unit of analysis.

[17] The Chinese court system is structured across four levels: county-level primary courts (over 3000), city-level intermediate courts (over 400), province-level high courts (31), and one supreme court. All courts may function as first-instance courts but handle various types of criminal cases, with lower-level people's courts supervised by higher-level courts.

[18] Appendix Figure A.7 presents the time trends of these crimes by province.

[19] We use the 2010 population as the denominator because it reflects the resident population more accurately than the Hukou population measure, providing a more precise physical representation of the populace.

[20] Our analysis focuses on procurement contracts for surveillance cameras issued by prefecture-level governments.



(a) Major Crimes (b) Other Crimes

Fig. 2. Crime rates by type at the national level, 2014–2019. Notes: Crime rate is measured by cases per 10,000 residents. Property_crime represents the total number of four types of criminal cases: robbery, stealing, defraudation, and extortion. Eco_crime denotes the total number of criminal cases related to finance and intellectual property infringement.

Other individual-level variables. In addition to examining crime rates, our study explores the impact of surveillance camera installation on individuals' satisfaction with the government, the frequency of outdoor exercises, hours worked, and their sense of security. We use three sets of micro-level survey data for this analysis. The China Family Panel Studies (CFPS) from 2014 to 2020 assess attitudes toward the government and the frequency of exercises. The China Labor-force Dynamics Survey (CLDS) from 2014 to 2018 provides data on security and hours worked. The World Values Survey (WVS) from 2012 and 2018 examines feelings of safety, confidence in the police, and experiences of crime. Detailed descriptions of these surveys are in Appendix D. The summary statistics of the key variables are provided in Appendix Table A.1.

3.2. Descriptive statistics

Fig. 2 offers a detailed analysis of crime rates by type from 2014 to 2019, with rates measured per 10,000 residents. Initially, many crime categories showed stable or slightly increasing trends up to 2018. This year marked a significant turning point due to the nationwide implementation of the *Skynet* Project, after which most crime rates, including traffic felonies, violence, drugs, and a combined category of robbery, stealing, defraudation, and extortion (labeled as "property crime"), began to exhibit sharp declines. In particular, property crimes decreased by 37.33% from 1.50 to 0.94 per 10,000 residents in a single year; violence crimes decreased by 38.38% from 0.99 to 0.61, and drug-related crimes fell by 47.92% from 0.48 to 0.25. These substantial reductions coincided with the operational period of *Skynet* Project, as illustrated in Fig. 1.

Although the simultaneous occurrence of two events does not establish causality, our study aims to investigate whether the extensive installation of surveillance cameras has causally affected the reductions in crime rates during our study period. We exploit variations in both surveillance camera installation and crime rates at the prefecture level. Panel (a) of Fig. 3 illustrates the geographic distribution of our key independent variable: cumulative per capita purchases of surveillance cameras from 2014 to 2019. Notably, installation intensity is exceptionally high in Beijing and its surrounding regions, the Southeast Coastal areas, and regions with significant minority populations, such as Guangxi Province, Inner Mongolia, and southern Tibet. Panel (b) presents the prefecture-level crime rates in 2014. Visually, there appears to be a positive correlation between crime rates in 2014 and the intensity of surveillance camera investment during the period 2014 to 2019 - a relationship that we later confirm statistically. Furthermore, panel (c) suggests that changes in crime rates from 2014 to 2019 are negatively correlated with the intensity of surveillance camera investment during the same period. Fig. 4 plots the correlation between the growth rate of crime from 2014 to 2019 and the cumulative purchases of surveillance cameras during this period.[22] Each dot represents a prefecture, clearly indicating a negative relationship between these variables, consistent with the visual patterns observed in panels (a) and (c) of Fig. 3.

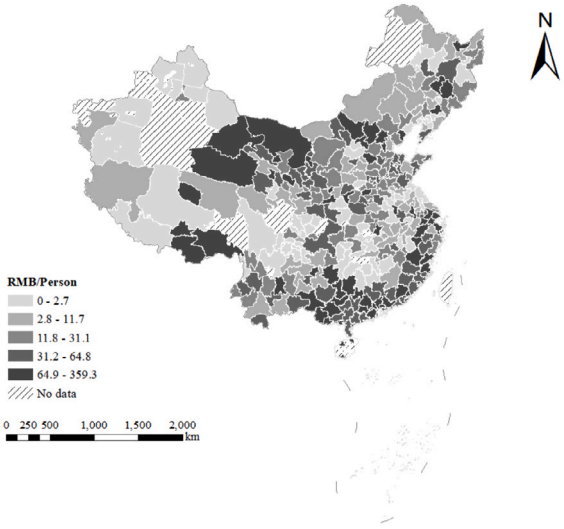
We intend to conduct a prefecture-level regression analysis to examine the relationship between changes in crime rates from 2014 to 2019 and the level of investment in surveillance cameras over the same period. The challenge in this analysis is the endogeneity of our key explanatory variable: Cities with higher investments in surveillance cameras may differ in unobserved ways from those with fewer investments, potentially influencing crime rate changes. Our approach takes three steps to address this endogeneity. First, we control for the main determinants that might have influenced the intensity of camera installation. Second, we use an instrumental variable (IV) strategy, employing the total number of camera firms or the total number of electronic firms existing before 2010 as the instrument for the cumulative investment in surveillance cameras between 2014 and 2019. Third, we conduct a series of placebo tests to confirm the validity of our IV strategy and reinforce the causal interpretation of our results.

3.3. Factors affecting surveillance cameras installation

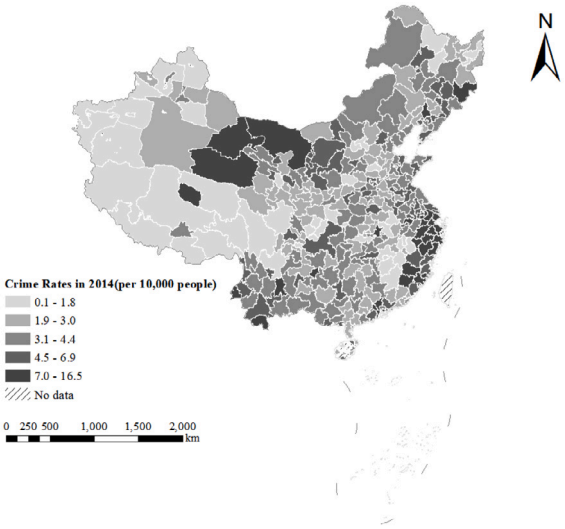
In Appendix Table A.4, we provide an analysis of the factors that influence the installation of surveillance cameras. We evaluate the intensity of surveillance camera installation using three metrics: the total number of government procurement contracts for surveillance cameras (columns (1) and (2)), the total number of government procurement contracts for surveillance cameras with price details (columns (3) and (4)), and the cumulative purchases (in RMB) of surveillance cameras (columns (5) and (6)), all expressed as the logarithm of the aggregate figures from 2014–2019. We focus on 12 city characteristics as potential influencers of surveillance camera installation intensity. Six of these 12 variables – initial GDP per capita, inequality (measured as

[21] Appendix C provides a detailed analysis of surveillance camera procurement patterns. Municipal governments in China award 50%–60% of contracts to firms within the same city and another 20%–30% to those within the same province. Out-of-province suppliers receive only 10%–20% of contracts, a share that has been declining over time.

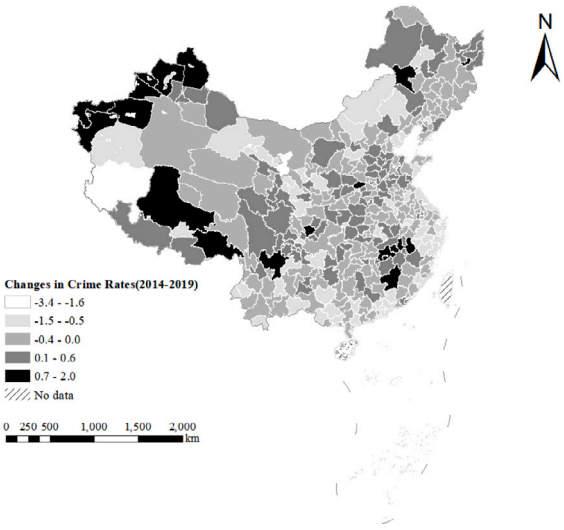
[22] For robustness, we use other measures of camera investment to plot the correlation between the growth rate of crime from 2014 to 2019 and the cumulative procurement amount of surveillance cameras during this period, as shown in Appendix Figure A.2.



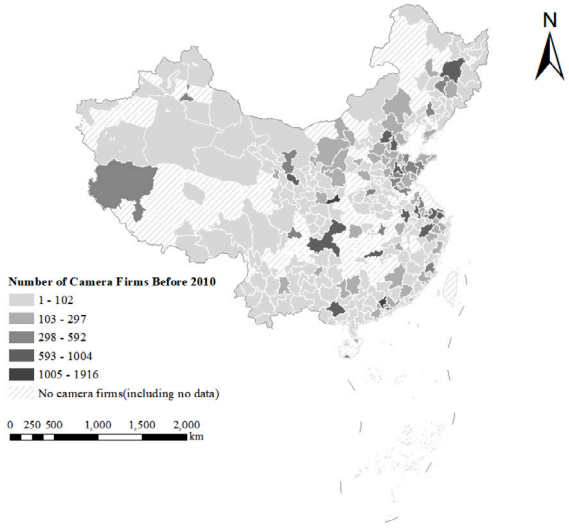
(a) Cumulative Purchases of Surveillance Cameras per capita from 2014 to 2019



(b) Crime Rates in 2014



(c) Changes in Crime Rates from 2014 to 2019



(d) The Number of Surveillance Camera Producers before 2010

Fig. 3. The geographic distribution of camera density, Crime rates, and Surveillance camera producers. *Notes:* Panel (a) illustrates the geographic distribution of cumulative purchases of surveillance cameras between 2014 and 2019. Panel (b) maps the geographic distribution of crime rates in 2014, providing a baseline snapshot of criminal activity prior to the intense deployment of cameras. Panel (c) shows the geographic distribution of the changes in crime rates from 2014 to 2019, highlighting regions where crime rates have increased or decreased during the period of increased surveillance. Panel (d) presents the geographic distribution of surveillance camera producers prior to 2010, providing insight into the industrial capacity of surveillance technology before its widespread adoption by government entities.

urban/rural per capita income), population, employment rate, domestic immigrant share, and sex ratio – follow (Edlund et al., 2013), who study how rising sex ratios contributed to the rising crime rate in China between 1988 and 2006. Two control variables – economic openness (measured by (Export+Import+FDI)/GDP) in 2014 and export growth between 2014 and 2019 – follow (Ma et al., 2023), who identify a causal relationship between export slowdown and crime rate in China during the 2010s. Additionally, we control for the initial crime rate, ethnic minority share, social security and employment expenditure per capita, and public security expenditure per capita.[23] In columns (1), (3), and (5), each explanatory variable is independently analyzed, and the results are pooled in one column, with each row representing a distinct regression. In columns (2), (4), and (6), all 12 city characteristics are included simultaneously as determinants of camera installation intensity.

As indicated by Appendix Table A.4, several factors significantly influence the installation of surveillance cameras. A higher initial crime rate, GDP per capita, economic openness, and fiscal expenditure on

[23] Controlling for the initial crime rate and public security expenditure per capita helps mitigate the potential estimation bias caused by omitting private camera installation. If high initial crime rates were the primary driver of private camera installations, this control would account for their effect. Additionally, since private demand for security cameras likely responds to gaps in public security, incorporating public security expenditure further alleviates concerns about missing the measure of private camera adoption.

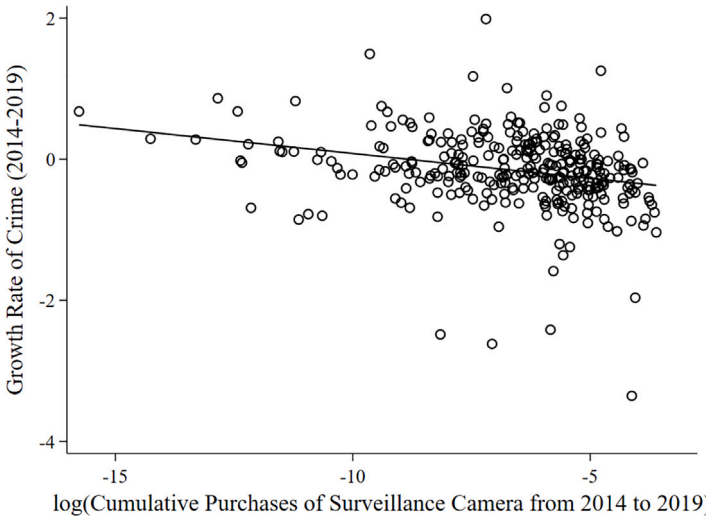


Fig. 4. The correlation between the growth rate of crime and the cumulative purchases of cameras, 2014–2019. *Notes:* Each dot in the figure represents a prefectural city. The growth rate of crime is quantified by the logarithmic difference in the total number of criminal cases between 2014 and 2019. The cumulative purchases of cameras are calculated by dividing the total procurement amount of public surveillance cameras by governments from 2014 to 2019 by the population figures from 2010. This method ensures that the measurements reflect the relative intensity of surveillance investments in relation to the city populations, providing a standardized basis for comparing the impact of surveillance across different regions.

public security per capita generally consistently predict a higher intensity of installation. In contrast, a higher employment rate tends to correlate with a lower intensity of installation. The effects of population size and minority share on installation intensity are variable, showing opposite effects in certain cases and proving to be occasionally insignificant. The impacts of inequality, domestic immigrant share, and sex ratio generally meet expectations in most specifications, although with a few deviations. Collectively, these observations suggest that the Chinese government is more likely to install surveillance cameras in wealthier cities and those with initially higher crime rates, reflecting a proactive stance against potential increases in crime. To address endogeneity concerns related to omitted variables, we incorporate all 12 observable characteristics into our subsequent empirical analysis. Furthermore, Appendix Table A.5 details robustness checks to ensure that our key findings remain valid when these control variables are added sequentially.

4. Empirical analysis

In this section, we outline our empirical strategy to investigate whether the observed decrease in crime growth rates can be attributed to the large-scale installation of surveillance cameras. We begin by introducing our empirical specification. Subsequently, we present the results of the OLS estimation, followed by the IV results. We also perform a heterogeneity analysis to examine variations in the impact across different settings and discuss the effects of surveillance camera installation on other related outcomes.

4.1. Empirical strategy

We use the following econometric specification to estimate the effect of camera installation on changes in crime rate:

$$\Delta \log\left(CRIME_c\right) = \beta_1 \log\left(CAMERA_c\right) + \beta_2 \log\left(CRIME_{c,2014}\right) + \Omega' X_c + \varepsilon_c \quad (1)$$

where $\Delta \log\left(CRIME_c\right)$ represents the log difference in the number of criminal cases per 10,000 residents in city c, and we estimate this model for the long interval between 2014 and 2019. Our primary explanatory variable, $\log\left(CAMERA_c\right)$, is the log of cumulative investment in surveillance cameras *per capita* in city c during the same period. As explained in the data section, we use three measures for this variable: first, the total number of procurement contracts for surveillance cameras; second, the number of contracts with price information; and third, the total expenditure on surveillance cameras. We adjust each measure by the city's population in 2010. The variable $\log\left(CRIME_{c,2014}\right)$ captures the initial crime rate in 2014, while X_c includes control variables such as population size, GDP per capita, inequality, employment rate, domestic immigrant share, minority share, and sex ratio, as discussed in Section 3.3.[24] In the main specification, we use a long difference specification to estimate the *cumulative* impact of camera investment on the five-year growth rate in criminal cases. The endogeneity issue is addressed by identifying an instrument for the time-invariant endogenous explanatory variable–cumulative investment in surveillance cameras per capita from 2014 to 2019. In Section 4.6, we exploit the panel structure of the data, which provides an estimate of the *average* effect of camera installation on crime rate reduction over 2014–2019. As we show later, the panel estimation effect is smaller than the long-difference estimator because the effect of the surveillance camera system escalates over time.

To address the endogeneity of $\log\left(CAMERA_c\right)$, we employ the number of camera firms in 2010 as an instrument for $\log\left(CAMERA_c\right)$. The rationale for this instrument is based on the presumption that the number of camera firms before 2010 influences the availability of local surveillance camera providers and, potentially, the financial costs associated with local governments procuring surveillance cameras in subsequent years. Therefore, this instrument is relevant. Furthermore, the pre-2010 count of camera firms is considered exogenous to subsequent local crime rate conditions. In the absence of the Chinese government's extensive surveillance initiative, the prosperity of the camera industry in a city would likely be independent of the local government's efforts and ability to combat crime, after accounting for common local characteristics affecting crime rates. The significance of the camera industry in 2010 lies in its facilitation of the extensive installation of surveillance cameras after the inception of the government project. Thus, this makes the pre-2010 camera firm count as a credible instrument. We first present the OLS results in Section 4.2, followed by the IV results in Section 4.3. We then provide evidence supporting the validity of our IV in Section 4.4, as well as evidence on the robustness of our IV results in Section 4.5.

[24] Appendix Table A.6 provides a detailed reference for these control variables.

Table 1
Impact of surveillance cameras on the growth rate of criminal cases (OLS).

Dep var.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	$\Delta \log CRIME_{c,2014-2019}$								
Measure of camera investment, 14-19:	log Cum. # of contracts	log Cum. # of contracts with price	log cumulative purchases	log Cum. # of contracts	log Cum. # of contracts with price	log cumulative purchases	log Cum. # of contracts	log Cum. # of contracts with price	log cumulative purchases
Camera investment, 14–19	−0.112***	−0.109***	−0.071***	−0.045**	−0.046**	−0.024*	−0.032	−0.032*	−0.015
	(0.020)	(0.021)	(0.014)	(0.018)	(0.018)	(0.013)	(0.020)	(0.019)	(0.014)
Observations	322	313	313	322	313	313	322	313	313
R-squared	0.091	0.086	0.070	0.244	0.234	0.228	0.322	0.312	0.309
Basic covariates	No	No	No	No	No	No	Yes	Yes	Yes
Initial crime rate in 2014	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates of the relationship between surveillance camera investment and changes in crime rates across Chinese cities from 2014 to 2019. The dependent variable is the log change in the number of criminal cases per capita over this period. Our key explanatory variable, labeled "Camera investment, 2014–2019", is captured using three alternative measures: the total number of procurement contracts for surveillance cameras (columns 1, 4, and 7), the number of contracts with price information (columns 2, 5, and 8), and the cumulative number of cameras purchased (columns 3, 6, and 9). All three measures are adjusted per 10,000 residents and expressed in logarithmic form. The regressions include a comprehensive set of controls: log population, GDP per capita, income inequality, employment rate, the share of domestic migrants, minority share, sex ratio, and a measure of economic openness (defined as the sum of exports, imports, and FDI divided by GDP). We also control for per capita fiscal expenditures on public security and on social security and employment, both measured in the initial year, as well as export growth over the sample period. Each observation represents a city. Robust standard errors are reported in parentheses, with significance levels indicated: *p < 0.10, **p < 0.05, ***p < 0.01.

4.2. OLS estimates

Table 1 reports the OLS estimation results of Eq. (1). As an initial step, columns (1)–(3) present the results without any control variables. In columns (4)–(6), we control for the initial crime rate only. In columns (7)–(9), we control for both the initial crime rate and the 12 covariates as explained in Section 3.3. We use the total number of procurement contracts for cameras (columns (1)(4)(7)), the total number of procurement contracts for cameras with price (columns (2)(5)(8)), and the cumulative purchases of cameras (columns (3)(6)(9)) as our measures of the investment in surveillance cameras.

We find that without the control variables, there is a significantly negative relationship between the intensity of surveillance camera installation and the crime growth rate in the three intensity measures (columns (1)–(3)), which is consistent with the graphical evidence in Fig. 4. This negative association becomes much smaller, yet still statistically significant, after controlling for the initial crime rate (columns (4)–(6)). This is because cameras were installed in cities with high-initial crime rates, and there is mean reversion in crime rate growth, so without cameras, these initially high-crime-rate cities would have experienced slow crime-rate growth anyway. Once this omitted variable is included in the regressions, cameras show a much weaker effect on crime growth. From columns (4)–(6) to (7)–(9), we further control for the 12 basic characteristics of the city related to crime. The coefficients further decrease and become insignificant across the three camera installation measures. The change in the coefficient of interest indicates the importance of controlling for the basic characteristics of the city related to crime, with the initial crime rate being the most critical control variable. Because the key explanatory variable, $\log\left(CAMERA_c\right)$, is endogenous, the OLS results should be interpreted with caution. In the following subsection, we turn to the instrumental variable approach.

4.3. IV estimates

We use a 2SLS approach for our IV results, estimating the following first-stage equation:

$$\log\left(CAMERA_c\right) = \beta asinh(CAMERA\ FIRMS_{c,2010}) + \gamma \log\left(CRIME_{c,2014}\right) + \tau X_c + \epsilon_c \quad (2)$$

where $\log\left(CAMERA_c\right)$ represents the cumulative investment in surveillance cameras per capita in city c during 2014–2019. Our instrumental variable, $CAMERA\ FIRMS_{c,2010}$, accounts for the number of firms producing surveillance cameras in the city c prior to 2010. An Inverse Hyperbolic Sine (IHS) transformation is applied (i.e., *asinh*) due to the presence of 64 cities without any camera firms before 2010. Among cities that had camera firms, the number ranged from 1 in Shuanghe, Xinjiang province, to 1916 in Xi'an, Shaanxi province, with an average of 144 camera firms per city. The variables $\log\left(CRIME_{c,2014}\right)$ and X_c are defined as previously in Eq. (1).

The bottom panel of Table 2 presents the first-stage estimation results (Eq. (2)), focusing on three distinct measures of camera investment employed in our main analysis. Each column corresponds to a different measure of government investment in surveillance cameras from 2014 to 2019. Across these measures, the variable $asinh(\#\ of\ camera\ firms)$ is significantly and positively associated with increased governmental procurement of surveillance cameras during this period, indicating that cities with a more developed camera industry in 2010 were more likely to engage in large-scale camera procurement in the following decade. The F-statistics for this instrument range from 64.13 to 212.96, well above conventional thresholds, affirming the strong predictive power of this instrument for our endogenous explanatory variable. Further evidence of the relevance of our instrument is provided in Appendix Figure A.3, where scatter plots and binned scatter plots illustrate a robust correlation and a log-linear relationship between the number of camera firms and the density of camera investment, further validating the instrument's effectiveness in our analysis.

The IV estimates in Table 2 reveal a significantly negative impact of camera installation intensity on changes in crime rates from 2014 to 2019. Columns (1)–(3) show IV estimates with the initial crime rate as the only control variable, while columns (4)–(6) include all control variables. Consistently across all columns, the installation of surveillance cameras significantly reduced crime growth rates. Column (6) – our preferred specification – suggests that a 100-log-point increase in cumulative surveillance camera purchases from 2014 to 2019 results in a 10.1% decrease in per capita criminal cases. Based on these results, Section 4.9 will provide a cost-effectiveness analysis, estimating criminal cases that are prevented and evaluating the economic costs. Finally, column (7) shows the reduced-form estimates, which indicate a significantly negative impact of the pre-2010 number of camera producers on changes in the crime rate from 2014 to 2019.

Comparing the OLS estimates from Table 1 with the IV estimates from Table 2, we observe that the IV coefficients are consistently larger in magnitude than the OLS coefficients in the corresponding columns: for example, −0.078 in column (1) of the IV results versus −0.045 in column (4) of the OLS results, −0.101 in column (6) of the IV results versus −0.015 in column (9) of the OLS results. This discrepancy suggests that the OLS results underestimate the impact of installing

Table 2
The impact of surveillance cameras on crime (Camera Firm IV).

Dep var.	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$\Delta \log CRIME_{c,2014-2019}$ IV						Reduced form
Camera investment, 14–19	−0.078*	−0.081*	−0.066*	−0.106**	−0.110**	−0.101**	
	(0.047)	(0.047)	(0.039)	(0.049)	(0.047)	(0.044)	
asinh(# of camera firms)							−0.047**
							(0.021)
Dep var.	First-Stage Regressions						
	log # of contracts	log # of contracts with price	log cumulative purchases	log # of contracts	log # of contracts with price	log cumulative purchases	
asinh(# of camera firms)	0.430***	0.431***	0.542***	0.435***	0.461***	0.505***	
	(0.026)	(0.027)	(0.048)	(0.031)	(0.032)	(0.064)	
Observations	322	313	313	322	313	313	322
Basic covariates	No	No	No	Yes	Yes	Yes	Yes
Initial crime rate in 2014	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First stage F-stat	165.72	153.85	85.26	188.72	212.96	64.13	

Notes: The top panel presents the IV estimates of the impact of camera investment on the crime growth rate, while the bottom panel shows the first-stage results, with each observation representing a city. The key regressor, representing the investment in surveillance cameras between 2014 and 2019, is measured by the total number of procurement contracts for cameras from 2014 to 2019 (columns 1, 4), the total number of procurement contracts with price information for cameras over the same period (columns 2, 5), and the cumulative purchases of cameras (columns 3, 6)–all adjusted per 10,000 residents and logarithmically transformed. In the top panel, column (7) displays the reduced-form result, using the logarithmic difference in the number of criminal cases between 2014 and 2019 as the dependent variable. In the bottom panel, the explanatory variable is inverse hyperbolic sine (IHS) of the number of camera firms before 2010. Basic covariates are described in Section 3.3. Robust standard errors are provided in parentheses, with significance levels indicated: *p < 0.10, **p < 0.05, ***p < 0.01.

surveillance cameras on the reduction of crime rates. This underestimation is anticipated: as illustrated in Appendix Table A.4, the Chinese government tends to install more cameras per capita in cities projected to experience a higher crime rate growth. Consequently, the intensity of the camera installation is positively correlated with the error term in Eq. (1), resulting in less negative OLS coefficients compared to the IV coefficients. This direction of bias in OLS is analogous to documented biases in studies on the effects of police on crime, where the literature often shows no effect or a positive effect of increasing the police force on crime rates. However, these effects generally become negative when endogeneity issues are addressed (Levitt, 1997; McCrary, 2002; Levitt, 2002; Tella and Schargrodsky, 2004; Gould and Stecklov, 2009; Machin and Marie, 2011; Draca et al., 2011a; Bove and Gavrilova, 2017; Chalfin and McCrary, 2018; Mello, 2019; Blesse and Diegmann, 2022).

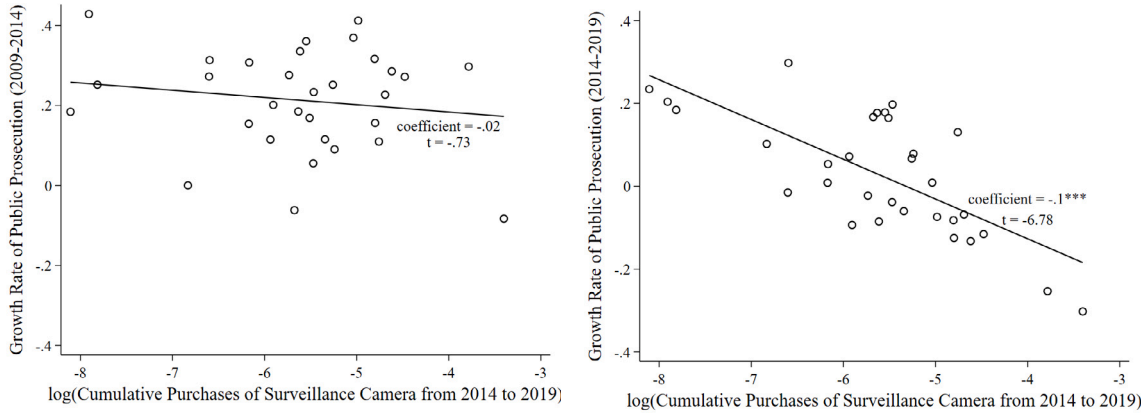
4.4. Discussion of instrument validity

The credibility of our IV results depends on the assumption that the number of camera firms before 2010 was not related to crime rate growth, once common factors are considered. This variable became relevant only after the Chinese government launched its public surveillance projects, which facilitated local surveillance and crime control. We present several pieces of evidence to support this assumption.

To validate our assumption, we collect crime rate data prior to 2014. Given the absence of data on city-level criminal cases before 2014, we collect province-level data on criminal cases from 2009 to 2019 from the *Procuratorial Yearbooks of China*. We specifically use the number of public prosecutions and the number of arrest approvals per 10,000 residents as proxies for crime rates, and then calculate the growth in these crime rate measures during the pre-treatment period (2009–2014) and the treatment period (2014–2019). Although this approach reduces our statistical power due to a smaller sample size, it enables us to investigate the growth of the crime rates prior to the widespread installation of surveillance cameras and perform placebo tests.

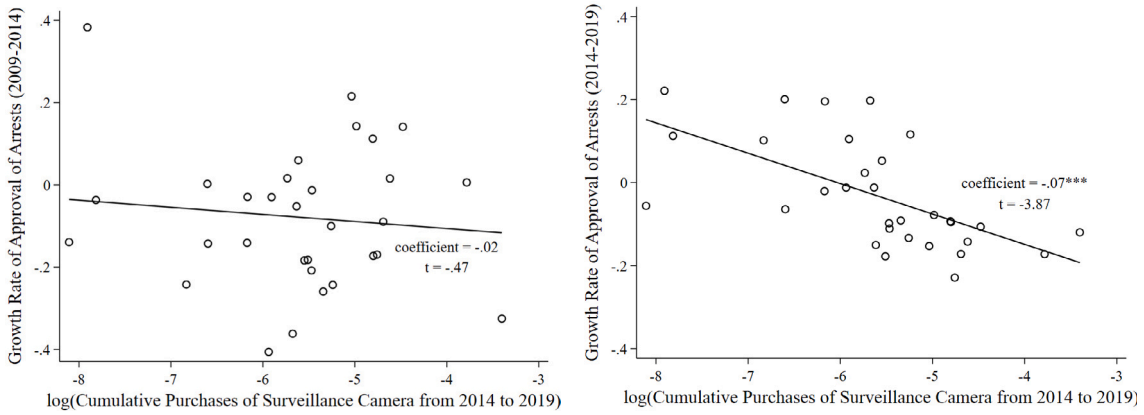
We begin by presenting simple scatter plots depicting the relationship between our key explanatory variable—the log of cumulative camera purchases per capita from 2014 to 2019—and our two outcome variables—the number of public prosecutions per capita (panels (a) and (b)) and the number of arrest approvals per capita (panels (c) and (d)). Panels (a) and (c) illustrate crime rate growth between 2009 and 2014, while panels (b) and (d) depict crime rate growth between 2014 and 2019. Each dot represents a province. Two notable patterns emerge: first, there appears to be a near-zero correlation between crime rate growth from 2009 to 2014 and the intensity of camera installation from 2014 to 2019, as indicated by both crime rate measures (panels (a) and (c)). Second, a significantly negative relationship is observed between the growth of the crime rate and the intensity of camera installation in the treatment period 2014–2019 (panels (b) and (d)). In general, these findings suggest that the intensity of camera installation from 2014 to 2019 did not specifically target provinces experiencing a faster or slower growth in crime rate during 2009–2014. However, after the government's use of surveillance cameras to combat crime, a negative association between the installation of surveillance cameras and the growth of crime rates emerged.

We use OLS regressions to validate the patterns observed in Fig. 5, with detailed results reported in Table 3. Different from the scatter plots, these regressions control for basic covariates and consider additional measures of camera investment. Panel A presents the baseline results, indicating the change in crime rate in 2014–2019, while panel B reports placebo tests with lagged results, reflecting the change in crime rate in 2009–2014. In columns (1)–(3), the dependent variable, crime rate, is measured by public prosecutions per capita, while in columns (4)–(6), it is measured by approvals of arrest per capita. The OLS results confirm the patterns observed in the scatter plots: during the 2014–2019 period, all three measures of camera installation intensity exhibit negative associations with the two measures of crime rate growth. Except for column (4), where the outcome is measured by approval of arrests and the explanatory variable by the log of the total number of camera contracts, all other five columns display statistically significant coefficients. In contrast, when the outcome is measured by the growth of crime rates between 2009 and 2014, across the three intensity measures of camera installation (still measured between 2014 and 2019) and the two crime rate measures, all coefficients are economically small and statistically insignificant. Therefore, this evidence further corroborates that the installation of surveillance cameras was uncorrelated with pre-trends in crime rates but began exhibiting a negative correlation after the government employed surveillance cameras to address crime.



(a) Public Prosecution (2009-2014) and Camera Investment (2014-2019)

(b) Public Prosecution (2014-2019) and Camera Investment (2014-2019)



(c) Approval of Arrests (2009-2014) and Camera Investment (2014-2019)

(d) Approval of Arrests (2014-2019) and Camera Investment (2014-2019)

Fig. 5. Test of pre-trends. *Notes:* Each dot represents a province in the scatter plots. The horizontal axis in all scatter plots represents the log of cumulative camera purchases from 2014 to 2019. Scatter plots (a) and (b) measure the outcome variable using the growth rate of public prosecutions, while scatter plots (c) and (d) use the growth rate of arrest approvals as the outcome variable. Scatter plots (a) and (c) function as placebo tests, assessing the growth rate of crime during the pre-treatment period (2009–2014). Scatter plots (b) and (d) examine whether there exists a significant negative association between camera investment and crime growth during the treatment period (2014–2019).

The previous analysis tests pre-trends based on the correlation between the explanatory variable and the outcome variable. To further bolster the credibility of IV results, we directly examine pre-trends by assessing the correlation between the instrumental variable and the outcome variable. The findings are reported in Table 4. In panel (a), changes in crime rates are regressed on the instrument, representing the reduced-form results. In panel (b), IV results are reported using the log of the number of camera firms as the instrument for the log of cumulative purchases of cameras.[25] Columns (1) and (2) measure the changes in the crime rate over 2014–2019 (by public prosecution and approval of arrest, respectively). Across both panels, we observe negative coefficients in these two columns, indicating that the log of the number of camera firms before 2010 predicts a slower growth rate in crime during 2014–2019. In contrast, columns (3) and (4) consider the changes in crime rates from 2009 to 2014 (lagged outcomes). We find positive coefficients in the two panels, but none are statistically significant. In summary, the results in Table 4 align with our central assumption: the instrumental variable – the number of camera firms before 2010 – was irrelevant to crime rate growth before 2014, but it became relevant afterward, coinciding with the massive installation of surveillance cameras to monitor and trace criminal activities.

Furthermore, we examine whether the level of crime rate suddenly exhibited a more negative correlation with the log of the 2010 number of camera firms after 2014. As shown in Table A.9 in the appendix, the log of the 2010 number of camera firms was positively correlated (column (1)) or uncorrelated (column (2)) with the crime rate in the years before 2014. However, such an association significantly dropped after 2014. Figure A.4 in the appendix confirms this dynamic pattern. Taking all the above results together, we find substantial evidence that the number of preexisting local camera firms suddenly became relevant for both the levels and growth of crime rates after 2014, which supports our interpretation that a larger number of local camera firms facilitated the installation of surveillance cameras, thus aiding local law enforcement in combating crime in this period.

4.5. Robustness checks

We test the robustness of the IV results to alternative measures of the instrument and alternative measures of camera investment. First, instead of using the baseline IV —$asinh\left(CAMERA\ FIRMS_{c,2010}\right)$—to measure the local availability of camera producers, we consider a measure of market access to nationwide camera producers where we weight each camera producer by its distance to the specific city. Specifically,

[25] In practice, we opt for the log transformation instead of the IHS transformation due to the use of provinces as units of analysis, mitigating issues with zeros in x.

Table 3
The impact of surveillance cameras on crime—province-level placebo test.

Panel A. Baseline results at the province level, 2014–2019						
Dep var.	Δlog(*CRIME*), 2014–2019					
	Public prosecution			Approval of arrest		
	(1)	(2)	(3)	(4)	(5)	(6)
log(Cum. # of contracts for cameras)	−0.108***			−0.051		
	(0.019)			(0.031)		
log(Cum. # of contracts with price for cameras)		−0.104***			−0.052*	
		(0.018)			(0.028)	
log(Cumulative purchases of cameras)			−0.089***			−0.057*
			(0.024)			(0.030)
Observations	31	31	31	31	31	31
R-squared	0.662	0.680	0.590	0.432	0.451	0.471
Basic covariates	Yes	Yes	Yes	Yes	Yes	Yes
Public prosecution in 2014	Yes	Yes	Yes			
Approval of arrest in 2014				Yes	Yes	Yes
Panel B. Placebo test with lagged outcomes, 2009–2014						
Dep var.	Δlog(*CRIME*), 2009–2014					
	Public prosecution			Approval of arrest		
	(1)	(2)	(3)	(4)	(5)	(6)
log(Cum. # of contracts for cameras)	−0.010			0.010		
	(0.029)			(0.045)		
log(Cum. # of contracts with price for cameras)		−0.006			0.008	
		(0.029)			(0.044)	
log(Cumulative purchases of cameras)			−0.032			−0.053
			(0.032)			(0.048)
Observations	31	31	31	31	31	31
R-squared	0.198	0.196	0.232	0.259	0.259	0.306
Basic covariates	Yes	Yes	Yes	Yes	Yes	Yes
Public prosecution in 2009	Yes	Yes	Yes			
Approval of arrest in 2009				Yes	Yes	Yes

Notes: Panel A presents the baseline results at the province level for the period 2014–2019. Panel B showcases a placebo test with lagged outcomes for the period 2009–2014. Each observation represents a province, and estimation is conducted using OLS. In columns (1)–(3), the dependent variable is the log difference in the number of public prosecutions per capita, while in columns (4)–(6), it is the log difference in the number of approvals of arrest per capita. The key regressor is the level of investment in surveillance cameras between 2014 and 2019. We also control for the initial level of public prosecutions and approvals of arrest separately. Basic covariates include the log of population, log of GDP per capita, measures of inequality, employment rate, domestic immigrant share, minority share, and sex ratio. Robust standard errors are reported in parentheses, with significance levels denoted as follows: *p < 0.10, **p < 0.05, ***p < 0.01.

Table 4
Placebo tests for the instrument.

Dep var.	Δlog(*CRIME*), $t_0 - t_1$			
	2014–2019		2009–2014	
	Public prosecution (1)	Approval of arrest (2)	Public prosecution (3)	Approval of arrest (4)
Panel A. OLS				
log(Number of camera firms)	−0.068**	−0.054	0.014	0.031
	(0.033)	(0.034)	(0.047)	(0.054)
Observations	31	31	31	31
R-squared	0.420	0.401	0.198	0.267
Basic covariates	Yes	Yes	Yes	Yes
Initial prosecution rate	Yes		Yes	
Initial arrest rate		Yes		Yes
Panel B. IV				
log(Cumulative purchases of cameras)	−0.127**	−0.107*	0.027	0.064
	(0.055)	(0.065)	(0.075)	(0.102)
Observations	31	31	31	31
R-squared	0.546	0.379	0.105	0.066
Basic covariates	Yes	Yes	Yes	Yes
Initial prosecution rate	Yes		Yes	
Initial arrest rate		Yes		Yes
First stage F-stat	4.46	3.88	4.87	4.27

Notes: This table presents OLS estimates in Panel A and IV estimates in Panel B at the province level. The dependent variables are the logarithmic difference in the number of public prosecutions per capita in columns (1) and (3), and the log difference in the number of arrest approvals per capita in columns (2) and (4). In Panel A, the explanatory variable is the log of the number of camera firms, while in Panel B, it is the log of cumulative purchases of cameras from 2014 to 2019 (per 10,000 residents). Basic covariates include the log of population, the log of GDP per capita, measures of inequality, employment rate, the share of domestic immigrants, the share of minorities, and the sex ratio. Robust standard errors are reported in parentheses, with significance levels indicated as follows: *p < 0.10, **p < 0.05, ***p < 0.01.

we follow Head and Mayer (2006) and Hering and Poncet (2010) to measure the market access to camera firms across the country for city c according to $MA_c = \log(\sum_{c \neq d} D_{cd}^{-\theta} \times N_{d,2010} + D_{cc}^{-\theta} \times N_{c,2010})$, where $N_{d,2010}$ represents the number of firms producing cameras in city d that existed in 2010, D_{cd} denotes the distance in kilometers between city c and d, and the parameter θ denotes the elasticity of trade cost to distance. A city's distance to itself, D_{cc}, is approximated as the weighted average distance from the city center to all other points within the city, assuming that the city is circular.[26] Appendix Table A.7 presents the IV coefficients under this set of alternative instrument variables. Across different trade elasticities ($\theta = 4$ or $\theta = 5$) and different D_{cc} assumptions, the IV coefficients range from −0.087 to −0.165, which center around our baseline estimate −0.1. All IV estimates are significant at the conventional levels. Because there is no consensus on the most appropriate value for θ and D_{cc}, we prefer the local availability of camera producers as our baseline IV.

In a second robustness check, we consider the depreciation rate to refine our measure of camera investment. The literature has assumed that surveillance cameras need to be replaced every five years (Priks, 2015a), which is a very rapid depreciation rate. We consider four possible annual depreciation rates—0.1, 0.2, 0.3, and 0.4, respectively—to obtain a more precise measure of the effective investment in cameras made from 2014 to 2019. The resulting IV coefficients are reported in Appendix Table A.8, which range from −0.099 to −0.101. Therefore, taking into account the depreciation rate in measuring camera investment has little effect on our estimated results.

In a third robustness check, we consider an alternative IV for the installation of cameras: the number of electronic producers in 2010. The logic for this instrument is, on the one hand, having an existing local electronics industry will be correlated with the capacity to scale up camera production when future government procurement contracts for cameras arrive, so this is a relevant instrument; on the other hand, the presence of electronic firms is not directly related to crime rate growth, and it becomes indirectly related to crime rate growth only after massive surveillance projects were initiated. With this alternative instrument, we report the IV regression results parallel to Table 2 in Table A.10 in the appendix. Comparing these two sets of results, we find similar coefficients. The coefficients in columns (1)–(3) in Table A.10 are insignificant, which is probably due to the weaker predictive power of local electronic firms in the first-stage regressions. The similar magnitudes across these two sets of IV results bolster the key findings of our paper.

4.6. Timing of the treatment effect

Our baseline regressions use a long-difference specification, regressing changes in crime rate from 2014 to 2019 on cumulative investment in cameras per capita during the same period. In this subsection, we exploit the panel structure of our data set and estimate the following econometric model:

$$\Delta \log CRIME_{c,2014-t} = \beta_1 \log Camera_{c,2014-t} + \sum_t X_{c,t0} * \delta_t + \sum_t \delta_t + \epsilon_{ct} \quad (3)$$

where $\Delta \log CRIME_{c,2014-t}$ is the log change in crime rate between 2014 and year t, the explanatory variable $Camera_{c,2014-t}$ is the cumulative investment in cameras per capita between 2014 and year t, $t \in \{2015, 2019\}$. We define the explanatory variable and outcome variable in this way instead of in yearly changes because installation of surveillance cameras per capita is a stock variable, so yearly changes in camera stock will miss the effect of pre-existing camera stock on crime rates. δ_t is the year dummy, and $X_{c,t0} * \delta_t$ is the set of interaction terms between initial city characteristics and year dummies. β_1 is the coefficient of interest. We do not include city fixed effects in Eq. (3) because the instrument we use – the number of local camera firms in 2010 – is a static variable, which has no explanatory power in the first stage once city fixed effects are included.

In the first stage, we regress

$$\log Camera_{c,2014-t} = \sum_t \beta_{1,t} CameraFirms_{c,2010} * \delta_t + \sum_t X_{c,t0} * \delta_t + \sum_t \delta_t + \epsilon_{ct} \quad (4)$$

That is, we interact the time-invariant instrument, $CameraFirms_{c,2010}$, with year dummies to predict the cumulative stock of cameras in every city in every year.[27]

Table A.12 reports the IV regression results under specification (3) and (4). Columns (1)–(3) in Table A.12 are comparable to columns (4)–(6) in Table 2. They differ only in whether using a panel structure or a long-difference structure. As we can see, the three coefficients are −0.047, −0.05, and −0.044 under the panel structure, while they are −0.106, −0.110, and −0.101 under the long-difference structure, all statistically significant at the 5% level, suggesting that an increase in camera investment leads to a significant decrease in crime rate. A larger effect obtained from the long-difference model estimation is consistent with our expectations because, as we reviewed in the policy background and in Figure A.1 in the appendix, it is in the year 2019 when we saw much more frequent media coverage of using surveillance cameras to combat crime. So 2014–2019 allows for a long enough time period for this infrastructure intervention to take effect.

To further verify our conjecture, we estimate three long-difference models, defining the changes over 2014–2017 (columns (1)–(3)),2014–2018 (columns (4)–(6)), and 2014–2019 (our baseline, columns (7)–(9)), respectively. Table A.13 reports such results. From column (1) to (9), we see that the crime-reduction effect increases sharply as we expand the time window. This result verifies our conjecture that the effect of cameras is strongest when a nationwide system is implemented (consistent with the name of the project, "Skynet," which means a massive net that no criminal can escape).

4.7. Heterogeneity analysis

The granularity of our data enables us to explore the heterogeneous effects of the installation of surveillance cameras on crime rates. In particular, we investigate the impact of surveillance cameras on changes in crime rates by type. We consider ten types of crime: public security-related crimes, traffic felony, violent crimes, property crimes (the sum of robbery, stealing, defraudation, and extortion), drug-related, prostitution, gambling, economic crime, counterfeiting, and bribery, with their effects reported in columns (1)–(10) of Table 5, respectively. To maintain conciseness, we present only IV results with the full set of controls and use the cumulative purchase of cameras as the measure of camera investment intensity, corresponding to column (6) of Table 2.

Table 5 reveals a substantial and statistically significant effect of the installation of surveillance cameras on various types of crime. Particularly noteworthy are the impacts on public security-related crime, violent crime, property crime (comprising robbery, stealing, defraudation, and extortion), and drug-related crime. The estimated coefficients suggest that a 100-log-point increase in cumulative purchases of cameras corresponds to a 14.3% reduction in drug-related crime, a 12.9% reduction in property crime, a 9.1% reduction in public security-related crime, and a 6.8% reduction in violent crime. In contrast, no discernible effects are observed on traffic felonies, prostitution, gambling, counterfeiting, or other economic crimes. A plausible explanation for this lack

[26] We set D_{cc} to be one-quarter or one-half of the city radius R, where the appropriate value depends on how concentrated the economic activity is in the prefectural city center. In addition to $R/4$ and $R/2$, we also set D_{cc} to 20 km (the average radius of Chinese prefectural cities is 71.6 km).

[27] The pre-2014 camera stock is small and almost negligible, so the 2014-to-present change is almost equal to the stock.

Table 5
Heterogeneous impacts across types of crime.

Dep var.	Δlog (CRIME), 2014–2019				
Crime type:	Public security	Traffic felony	Violence	Property	Drugs
	(1)	(2)	(3)	(4)	(5)
log(Cumulative purchases of cameras)	−0.091**	−0.038	−0.068*	−0.129***	−0.143***
	(0.038)	(0.042)	(0.040)	(0.042)	(0.038)
Observations	293	311	312	312	302
Basic covariates	Yes	Yes	Yes	Yes	Yes
Crime rate in 2014	Yes	Yes	Yes	Yes	Yes
First stage F-stat	76.91	73.00	73.57	76.14	86.97
Crime type:	Prostitution	Gambling	Econ crime	Counterfeiting	Bribery
	(6)	(7)	(8)	(9)	(10)
log(Cumulative purchases of cameras)	−0.075	−0.009	−0.017	−0.013	−0.073*
	(0.064)	(0.038)	(0.036)	(0.053)	(0.044)
Observations	255	279	250	239	284
Basic covariates	Yes	Yes	Yes	Yes	Yes
Crime rate in 2014	Yes	Yes	Yes	Yes	Yes
First stage F-stat	109.70	80.63	76.81	56.54	67.98

Notes: This table reports IV estimates of the effect of surveillance cameras on different types of crime. Each observation is a city. The dependent variable is the 2014–2019 log difference in the total number of corresponding criminal cases per 10,000 residents. The explanatory variable is the log of cumulative purchases of cameras from 2014 to 2019 (per 10,000 residents). Basic covariates are described in Section 3.3. Robust standard errors are enclosed in parentheses, with significance levels noted: *p < 0.10, **p < 0.05, ***p < 0.01. Property_crime represents four types of crime: robbery, stealing, defraudation, and extortion. Eco_crime denotes the crime related to finance and intellectual property infringement.

Table 6
Heterogeneous impacts by regional and offenders' characteristics.

Dep var.	(1)	(2)	(3)	(4)	(5)	(6)
	Log change in criminal cases per capita ($\Delta \log CRIME_{c,2014-2019}$)					
Group variable	% Migrants > median	% Minority > median	% Male > median	GDP per capita > median	Offenders Aged > 44	Offenders Female
log(Cumulative purchases of cameras)	−0.062	−0.040	−0.016	−0.079*	−0.065	−0.059*
	(0.041)	(0.037)	(0.034)	(0.043)	(0.046)	(0.031)
log(Cumulative purchases of cameras)	−0.057	−0.222*	−0.142**	0.053	−0.112**	−0.042
×Group Indicator	(0.076)	(0.115)	(0.059)	(0.051)	(0.045)	(0.031)
Group Indicator	−0.289	−1.518*	−1.031**	0.393	−0.758***	−1.957***
	(0.467)	(0.794)	(0.408)	(0.364)	(0.129)	(0.147)
Observations	313	313	313	306	620	608
Basic covariates	Yes	Yes	Yes	Yes	Yes	Yes
Crime rate in 2014	Yes	Yes	Yes	Yes	Yes	Yes
First stage F-stat	11.45	13.92	35.66	23.59	63.61	66.33

Notes: This table presents IV estimates of the varied impacts of camera installation on crime, categorized by regional and offenders' characteristics. Each entry represents a city. The dependent variable is the 2014–2019 log difference in the total number of criminal cases per capita, while the explanatory variable is the cumulative purchases of cameras from 2014 to 2019 (logged, per 10,000 residents). Each observation is a city in columns (1)–(4) and is a city-demographic group pair in columns (5) and (6). Basic covariates are described in Section 3.3. Robust standard errors are provided in parentheses, with significance levels indicated as follows: *p < 0.10, **p < 0.05, ***p < 0.01.

of effect is the inherent difficulty in detecting these types of criminal activity through surveillance cameras. Collectively, these findings suggest that the Chinese police system has effectively utilized cameras to solve criminal cases, thereby exerting a strong deterrent effect on criminal activities more susceptible to exposure by public cameras, consequently leading to a significant reduction in the respective crime rates.

We proceed to investigate the heterogeneous effects of surveillance camera installation on crime by city or offender characteristics in Table 6. In particular, we focus on how the impact differs based on the share of domestic immigrants, the share of minorities, the sex ratio, GDP per capita, and offenders' age and gender. For brevity, we only report the IV results with the full set of controls and only use cumulative purchases of cameras as the measure of camera investment intensity. For each heterogeneity variable, we construct a dummy indicating whether the observation is above the median value or not, and we then interact this group indicator with our key explanatory variable. The results in columns (1)–(4) demonstrate that the crime-reduction effect of surveillance cameras is significantly larger in cities with an above-median share of minorities and in cities with an above-median share of males. The effect is also larger in cities with an above-median share of migrants and in cities with a below-median GDP per capita, but such a distinction is not statistically significant. This heterogeneity is expected because these cities presumably have higher crime rate growth potential, even after controlling for the characteristics, so surveillance technology plays a more important role there. Furthermore, in columns (5) and (6), we break down criminal cases by age group and gender of offenders and report the effect of surveillance camera installation on crimes committed by these different groups. We find that the effect is stronger for criminal cases committed by those aged above 45, but there are no significantly differential effects on crimes committed by males or females. A possible interpretation is that surveillance cameras had a more significant deterrent effect on older potential offenders.

4.8. Other outcomes

We next examine whether the widespread public surveillance system has impacted individuals' subjective feelings, such as their sense of security and satisfaction with the government, as well as their behaviors, including the hours they work and the time they spend outdoors. This analysis serves two main purposes. First, it provides empirical insights into whether a decline in crime rates translates into improvements in people's well-being, which is equally important as the

Table 7
The impact of surveillance cameras on sense of security, labor supply, frequency of exercise, and government satisfaction.

Dep var.	(1) Sense of security Δ 2014–2019	(2) Labor supply Δ 2014–2019	(3) Frequency of exercise Δ 2014–2019	(4) Government satisfaction Δ 2014–2019
Panel A: Overall effects				
log(Cumulative purchases of cameras)	0.023	0.063***	0.046	0.081***
	(0.017)	(0.023)	(0.081)	(0.031)
Observations	2197	2197	3306	3306
First stage F-stat	23.67	23.63	30.42	30.39
Panel B: Effects on female				
log(Cumulative purchases of cameras)	0.024	0.056**	0.031	0.068
	(0.018)	(0.024)	(0.096)	(0.042)
log(Cumulative purchases of cameras) × Female	−0.002	0.018	0.042	0.030
	(0.016)	(0.024)	(0.121)	(0.053)
Observations	2190	2190	3300	3300
First stage F-stat	21.81	21.45	22.55	22.4
Basic covariates	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes
Crime rate in 2014	Yes	Yes	Yes	Yes
Initial level of outcome	Yes	Yes	Yes	Yes

Notes: This table presents IV estimates of the impact of camera investment on individuals' sense of security, hours worked, frequency of exercise, and satisfaction with the government. Each entry represents an individual. The outcome variables are calculated based on a balanced panel of individuals and then taken first (log) differences. The dependent variables include changes in the sense of security and hours worked from 2014 to 2018 (columns (1)–(2), from CLDS), changes in the frequency of exercise and satisfaction with the government from 2014 to 2020 (columns (3)–(4), from CFPS). The explanatory variable is cumulative camera purchases from 2014 to 2019 (logged, per 10,000 residents) in the individual's prefectural city of residence. Basic covariates are described in Section 3.3. Individual controls include gender, age, ethnicity, marital status, health status, and educational background. The initial level of the outcome variable is also controlled for. All control variables are also interacted with gender, so we cannot directly interpret the coefficient of the female dummy. Therefore, we do not report this coefficient in the table. Standard errors are clustered at the city level. Significant at *10%, **5%, ***1%.

objective measure of crime rates. Second, observing any effects on well-being and time allocation would reinforce the validity of our baseline findings regarding crime rate reduction.

For this analysis, we merge our camera installation data with two nationally representative individual-level datasets: the China Labor-force Dynamics Survey in 2014 and 2018, which provides information on subjects' sense of security and hours worked, and the China Family Panel Studies in 2014 and 2020, which includes subjects' satisfaction with the government and reported frequency of outdoor exercises. Utilizing balanced panels of individuals from both datasets, we estimate the impact of camera installation in local cities on these four individual outcomes.

The findings are presented in Table 7. Columns (1)–(4) report the impact of surveillance camera investment on individuals' sense of security, hours worked, frequency of outdoor exercise, and satisfaction with the government, respectively. Panel A estimates the effect on the overall population, while panel B estimates the heterogeneous effects by gender. Across all the columns and the panels, IV results with the full set of control variables are reported. The full sample analysis (panel A) reveals an enhancement in the sense of security (column (1)), indicating a 0.023 point (on a scale of 1–4) increase in safety perception with a 100-log-point rise in camera purchases, but such an effect is not significant. Then, we explore whether camera coverage impacted their labor supply patterns and time spent on outdoor activities. It is reasonable to conjecture that with reduced safety concerns, individuals may feel more comfortable working longer hours and spend longer time doing outdoor exercises, particularly at night. Columns (2) and (3) generally confirm this prediction: Column (2) demonstrates a significant increase in hours worked, suggesting that a worker on average worked for 6.3% longer hours following a 100-log-point increase in camera purchases. Column (3) reveals an increase in the frequency of outdoor exercises, though the effect is imprecisely estimated. Finally, consistent with these positive impacts, column (4) suggests that the individuals experienced a significant increase in satisfaction with the government (0.081 points on a 1–5 scale) in cities with 100-log-point higher per capita camera investment.

In panel B of Table 7, we investigate whether there are important gender differences in subjective feelings and behaviors following the widespread installation of cameras. Making this distinction is important because females are generally more susceptible to violent crimes.[28] Comparing the coefficients in these panels, we find a larger effect of camera installation on outcomes among women than men in terms of hours worked, frequency of our door exercise, and satisfaction with the government, and almost no differential effect on the sense of security. However, none of the gender differences are statistically significant.

We augment our analysis with data from the 2012 and 2018 World Values Survey (WVS) to delve deeper into the effects of surveillance camera installation on individuals' sense of security and attitudes toward the government and police. While the WVS offers a wider range of questions, it represents a repeated cross section of individuals. By aggregating individual results to the provincial level, we assess whether the presence of surveillance cameras enhanced people's sense of security. Detailed results are reported in the Appendix Table A.11. Our findings indicate a significant increase in the proportion of individuals who reported rare or no feelings of insecurity due to crime in provinces with higher camera density. Furthermore, we present national summary statistics depicting individuals' attitudes and self-reported crime experiences in the Appendix Figure A.6. These findings reveal a marked increase at the national level in the proportion of respondents who felt secure in their neighborhoods (a 20% increase), trusted the police (a nearly 20% increase), and experienced a decline in personal or family victimization rates over the past year (an 8% and 10% decrease, respectively). These statistics collectively suggest a substantial improvement in the sense of security and satisfaction with the government and police. Interestingly, the 2018 WVS data reveal

[28] For instance, it is reported that 86% of women in Brazil, 79% in India, and 86% in Thailand have experienced public harassment during their lifetime (Heath et al., 2024). Women are significantly more supportive of CCTV in public spaces than men and believe that CCTV reduces crime and increases feelings of safety (Hancock, 2004).

that more than 80% of Chinese citizens believed the government should have the undisputed right to deploy surveillance in public spaces, as shown in the Appendix Figure A.5.

In summary, the placebo results in Section 4.4, the robustness checks in Section 4.5, the heterogeneous effects in Section 4.7, and the impact on various outcomes in Section 4.8 collectively support the interpretation that the widespread deployment of surveillance cameras in China between 2014 and 2019 substantially reduced crime rates during the same period. However, there remains a possibility that conditional on fundamental city characteristics, our instrumental variable – the number of preexisting camera firms – might be correlated with unobserved factors, potentially biasing our IV estimates. However, any such bias should primarily affect crime types susceptible to public surveillance, but not those less detectable by cameras. Furthermore, if the crime reduction effect of surveillance cameras were absent, the observed enhancement in public security perception, governmental satisfaction, and increased hours worked in camera-intensive cities would need alternative explanations. We find this scenario improbable, given the concurrent nature of these phenomena.

4.9. Cost-effectiveness analysis

Our preferred IV estimates indicate that compared to a baseline city, criminal cases per capita decreased by an additional 10.1% over 2014–2019 in cities with 100-log-point more cumulative camera purchases per capita (column (6) in Table 2). Given the extensive investment in surveillance cameras and their nationwide deployment, this marginal effect could lead to a significant reduction in overall crime rates.

We perform a simplified cost-effectiveness analysis to estimate the cost of deterring one crime. Unlike previous studies that focused on local surveillance camera effects, our approach evaluates the impact of widespread camera deployment. Our econometric model employs the camera investment density to uncover a log-linear association between this density and changes in criminal cases from 2014 to 2019, as depicted in Fig. 4. This finding is significant, revealing diminishing returns to camera investment: initial investments yield the most significant crime reduction, while further investments become increasingly costly for similar crime reduction benefits due to the exponential growth of surveillance infrastructure.

Recognizing this log-linear relationship, we calculate the economic costs of deterring one crime under several counterfactual scenarios. It is worth pointing out that we only calculate the costs of deterring crime by cameras, but not the benefits of deterring crime, which could include avoidance of the physical, psychological, and financial harm incurred by crime victims, as well as savings in the financial costs of supporting law enforcement, the judiciary, and correctional services. We leave a careful calculation of these benefits of deterring crime in the Chinese context for future research. After excluding the top and bottom 1% of the observations to mitigate outlier effects, we assume that all 319 cities reduced their cumulative camera investment over the period 2014–2019 to the bottom 5th percentile, while cities below this threshold maintained their investment level.[29] This yields a remarkable increase of 281,189 criminal cases, with the counterfactual number in 2019 estimated at 731,433, against the actual 450,244 cases across these cities. We then calculate city-level costs by subtracting the minimum camera purchases from actual cumulative purchases, totaling 60.93 billion yuan (8.96 billion USD) across the 319 cities. Assuming a five-year camera replacement cycle (Priks, 2015a) and applying the 2019 effect over the next five years, we estimate the cost of deterring a criminal case by massive camera installation at 43,338 Yuan (6373 USD).[30]

In the next three counterfactuals, we assume that all cities with a camera investment density higher than the 10th, 20th, and 30th percentiles reduced their investment to the respective percentiles, while the remaining cities maintained their actual levels. In these scenarios, cities would collectively save 60.03, 57.35, and 53.86.6 billion RMB, respectively. However, the number of crimes would increase by 173,360, 104,650, and 72,141 cases in one year, resulting in average costs of reducing one crime at 69,252, 109,610, and 149,330 RMB (10,184, 16,119, and 21,960 USD), respectively, assuming the effect lasts for five years. These calculations highlight the diminishing marginal returns to camera investment, indicating the escalating unit costs of using surveillance for crime prevention as the surveillance network expands.

Comparison with existing literature. We compare our estimates with existing studies and summarize the results in Table 8. Our review identifies five closely related papers that analyze the cost-effectiveness of surveillance camera installation. These studies span five cities across four countries – the U.K., Sweden, Uruguay, and Colombia – covering contexts ranging from subway systems to city streets. Given differences in study settings, crime types, and methodologies, the estimated cost of preventing a single crime incident varies widely, from $583 per unruly incident in Sweden (Priks, 2014) to $38,214 per robbery incident in the U.K. Burrows (1980). The median cost ranges from $2000 per criminal case in Stockholm, Sweden (Priks, 2015a), to $6686 in Medellín, Colombia (Gómez et al., 2021), reflecting the substantial variation in estimates across different contexts. Our estimated cost – $6373 per criminal case in the extreme scenario where all cities had only a minimal number of cameras – closely aligns with the median estimate in the literature. The key distinction of our study lies in the geographic scope and scale of the surveillance project. Unlike previous studies, which primarily assess localized interventions targeting crime hotspots such as transportation hubs, our analysis examines a nationwide initiative involving at least 200 million cameras, with total expenditures reaching 11.3 billion USD by 2019. In comparison, the largest surveillance project among the studies reviewed had a total cost of just 5.61 million USD.

Given the diminishing crime-deterrent effect of installing surveillance cameras in less crime-prone areas and the vast scale of China's public surveillance project, one might anticipate a significantly higher average cost of deterring one criminal case in our context. However, our findings suggest that even on such a large scale, the per-unit cost of deterring crime remains within a reasonable range. This implies the economic viability of establishing a dense, nationally interconnected public surveillance system to combat crime. Our estimates of the costs of crime reduction through camera installation compare favorably with those in the U.S. and other contexts. For example, Miller et al. (2021) estimate the costs of crime by considering medical expenses, productivity losses, property damage, public services, legal proceedings, and penalties. They find that the weighted average costs for all crimes are $21,272 (in 2017 dollars), with the highest unit costs associated with homicide, rape, sexual assault, impaired driving crashes, child maltreatment, arson, physical assault, and robbery. Cornaglia et al. (2014) estimate that the society-wide impact of increasing the crime rate is about 80 times more than the direct impact on the victim in Australia. However, since we lack an estimate of crime costs in China, we refrain from directly comparing U.S. and Chinese contexts. Nevertheless, our estimates suggest diminishing returns as the surveillance network expands. The marginal costs of reducing additional crime can triple as the network's scale increases from the 5th percentile to the 30th percentile of all cities.

[29] We opt to reduce camera purchases to a positive number rather than zero to accommodate our log-based specification (Eq. (1)).

[30] This calculation has biases: it assumes consistent deterrence over the next five years, potentially exaggerated due to Covid-19 impacts in 2020–2022, and underestimates annual costs due to the inclusion of all surveillance system purchases, not just cameras, and lack of maintenance cost data.

Table 8
The estimated cost of reducing one crime by installing surveillance cameras in existing literature.

Country	Deployment area of cameras	Total costs	Types of cases covered	The cost of preventing one crime (incident)	Source
U.K	London underground	$155,532.80[a]	Theft and robbery	Theft:$1385.21; Robbery:$38,214.89	Burrows (1980)
Sweden	Stadiums	$169,672	Unruly behavior inside stadiums[b]	Unruly incident: $583.25	Priks (2014)
Sweden	Stockholm subway system	$5,000,000	Drug, assault, pickpocketing, violence against officials, robbery, etc.	Approximately $2000	Priks (2015a)
Uruguay	Montevideo	$350,000[c]	Theft, robbery, assault, domestic violence, damage, murder, etc.	$830	Munyo and Rossi (2020)
Colombia	Medellín	$4,480,000	Theft, robbery, homicide and assault	$6686	Gómez et al. (2021)

Notes:
[a] Installation and operating costs in the first year were £128,000 ($155,532.8) at 1975 prices.
[b] For instance, throwing objects such as coins, bottles, lighters, firecrackers, batteries, snuff boxes, etc.
[3] Considering that the average annual cost to operate each camera is $85,000, the total cost amounts to at least $85,000 per camera intersection, totaling $5,610,000 for 66 intersections.
[4] This translates to a reduction of approximately 29 injury accidents per year across all camera intersections in Houston, resulting in a cost of $5,610,000/29 = $193,448.3 per injury accident prevented.
[c] Furthermore, the total expenditure on citizen security and crime prevention in Uruguay amounts to 2.2 percent of GDP (or $1592).

5. Conclusion

China has established the world's largest video surveillance network, with an investment exceeding 76 billion yuan in surveillance cameras from 2002 to 2019. This study provides pioneering evidence for the causal relationship between this extensive surveillance and crime rates in China. Leveraging variations in the number of camera-producing firms before their widespread installation, we compile a unique and thorough city-level dataset encompassing criminal cases pre and post the large-scale camera deployment. In addition, we create a novel dataset on the procurement contracts for these cameras at the city level. By integrating these datasets with various micro surveys, we uncover broader impacts of surveillance cameras on social outcomes.

We demonstrate a notable reduction in crime rates in China associated with increased exposure to surveillance cameras. This effect is particularly pronounced in regions with a higher minority representation and greater gender imbalances. The effect is stronger on crimes related to drugs, robbery, theft, fraud, extortion, public security, and violence. We find that the widespread installation of cameras has improved people's sense of security and satisfaction with the government. An unintended consequence of this extensive surveillance deployment is a marked increase in hours worked, potentially attributed to heightened safety perceptions. Our cost-effectiveness analysis indicates that the economic viability of reducing one crime stands at approximately 6373 USD, compared to international benchmarks, albeit with rapidly increasing marginal costs.

The widespread acceptance of surveillance cameras in China, as evidenced by local support, suggests the effectiveness of such technologies. This underscores the importance of context in policymaking concerning surveillance and privacy. Policymakers must acknowledge the varying perceptions of human rights and privacy in diverse cultural and political contexts. Hence, international criticisms of surveillance practices should consider these nuances rather than applying a uniform standard. For nations considering similar surveillance measures, it is imperative to understand the perspectives of their citizens, ensuring that policies not only bolster public safety but also uphold societal values and expectations. Finally, there is a compelling need to establish a framework for a comprehensive analysis of the benefits and costs of the widespread deployment of surveillance cameras, a task we defer to future research endeavors.

CRediT authorship contribution statement

Hong Ma: Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization. **Mingzhi Xu:** Writing – review & editing, Writing – original draft, Validation, Funding acquisition, Formal analysis. **Wei You:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jinmei Feng:** Software, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jdeveco.2025.103557.

Data availability

Data will be made available on request.

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